

# **INSTITUTE OF ACTUARIES OF INDIA**

## **EXAMINATIONS**

**17<sup>th</sup> March 2017**

**Subject CT2 – Finance and Financial Reporting**

**Time allowed: Three Hours (10.30 – 13.30 Hours)**

**Total Marks: 100**

### **INSTRUCTIONS TO THE CANDIDATES**

- 1. Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
- 2. Mark allocations are shown in brackets.*
- 3. Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
- 4. Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

#### **AT THE END OF THE EXAMINATION**

**Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.**

- Q. 1)** Why does tax legislation not usually allow depreciation recorded in the books as an expense, but grants capital allowances instead for companies?
- A.** The tax structure is based on cashflow concept and not accrual basis of accounting.
  - B.** To bring consistency in tax assessment for all companies, it is important to grant capital allowances in a consistently defined way.
  - C.** Tax legislation does not cover expenses which are only estimated, such as depreciation.
  - D.** The Tax regulations do not consider depreciation as a revenue item available for deduction.
- [2]**
- Q. 2)** Which of the following best explains why a falling share price is thought to impose some discipline on weak directors of a company?
- A.** The shareholders and other investors will withdraw their money from the company
  - B.** The cost of capital will go up.
  - C.** The value of share options held by directors will decline
  - D.** The company will be vulnerable to a take-over.
- [2]**
- Q. 3)** ABC Ltd has 1,00,000 shares outstanding at a price of Rs 120 per share. A rights issue will allow 1 share for every 5 currently held by the shareholders for Rs 90 each. Which of the following is true?
- A.** The share price will fall to Rs 115 per share
  - B.** The company's total value will decrease to Rs 11,000,000
  - C.** The firm will raise Rs 9,00,000
  - D.** The number of shares outstanding will fall to 90,000
- [2]**
- Q. 4)** The following figures were taken from a company's accounts:
- |                  | 2016      | 2015     |
|------------------|-----------|----------|
| Operating Profit | 10,00,000 | 8,00,000 |
| Depreciation     | 1,00,000  | 1,20,000 |
| Working Capital  | 3,50,000  | 2,50,000 |
- (inventory + trade receivables – trade payables)
- What is the company's cash inflow from operating activities for the year ended 2016?
- A.** 12,00,000
  - B.** 11,00,000
  - C.** 9,00,000
  - D.** 10,00,000
- [2]**

- Q. 5)** A company purchased an item of inventory for Rs 1500 and sold it for Rs 2000. Due to rising prices, the inventory item's replacement cost is Rs 2100. Which of the following statements best describes the impact of the sale and replacement of this inventory?
- A.** The company will recognize a loss of Rs 100 under historical cost accounting, and has suffered a real loss of Rs 100.
  - B.** The company will recognize a loss of Rs 100 under historical cost accounting, and has made a real profit of Rs 500.
  - C.** The company will recognize a profit of Rs 500 under historical cost accounting, and has suffered a real loss of Rs 100.
  - D.** The company will recognize a profit of Rs 500 under historical cost accounting, and has made a real profit of Rs 500.
- [2]**
- Q. 6)** Which of the following is true regarding underwriting a security?
- I.** Underwriting means the financial intermediaries arrange to sell securities to the public or to other institutions for a fee
  - II.** The underwriters agree to take up any securities not sold at the pre-agreed price.
  - III.** Underwriting assures the entity requiring funds that the full amount of money sought will be obtained.
  - IV.** It creates confidence in the securities being issued.
- A.** I,II,III
  - B.** II,III
  - C.** I,II
  - D.** II,III,IV
  - E.** All of the above
- [2]**
- Q. 7)** A Company has issued the following:
- I.** Subordinated Loan stock
  - II.** Ordinary Shares
  - III.** Preference Shares
  - IV.** Floating Charge Debentures
- State the order of priority in the event of winding up of the business.
- A.** I,IV,III,II
  - B.** IV,I,III,II
  - C.** IV,I,II,III
  - D.** I,IV,II,III
  - E.** I,III,IV,II
- [2]**
- Q. 8)** A project having a life of one year is started with initial investment of 100 lakhs with 10% cost of capital. It's NPV and IRR is found to be 10 lakhs and 21%. If cost of capital is changed to 11% then which of the following is true?
- A.** NPV = 9 Lakhs and IRR = 21%
  - B.** NPV = 10 Lakhs and IRR = 23%
  - C.** NPV = 9 Lakhs and IRR = 23%
  - D.** NPV = 11 Lakhs and IRR = 21%
  - E.** NPV = 10 Lakhs and IRR = 21%
- [2]**

**Q. 9)** Agency theory identifies certain types of cost as agency costs. Which of the following would be an agency cost?

- I.** Directors' salaries
- II.** Directors' bonuses
- III.** The auditor's fee
- IV.** Employees' pay rises

- A.** I and II only
- B.** I, II and III only
- C.** II and III only
- D.** II, III and IV only
- E.** All of the above

[2]

**Q. 10)** Company ABC uses only debt and equity to finance its capital budget and uses CAPM to compute its cost of equity. The company estimates that its WACC is 12%. The capital structure is 75% debt and 25% equity. Before tax, the cost of debt is 12.5 % and tax rate is 20%. The risk free rate is 6% and the market risk premium is 8%. What is the beta of the company?

- A.** 1.5
- B.** 0.8
- C.** 1.67
- D.** 0.67
- E.** 1.8

[2]

**Q. 11)** Alpha Ltd. is a listed company manufacturing auto components. The management and directors are considering the draft financial statements for the year ended 31<sup>st</sup> December 2016 and these figures are not yet finalized.

The following is an analysis prepared by the accountant for discussion –

|                                            | Year ended 31 <sup>st</sup><br>December 2015 | Draft – Year ended 31 <sup>st</sup><br>December 2016 |
|--------------------------------------------|----------------------------------------------|------------------------------------------------------|
| Equity Shares (Rs 10 per share)            | 10,000,000                                   | 10,000,000                                           |
| EPS                                        | 1                                            | 1.2                                                  |
| Price/ Earnings Ratio                      | 13                                           | 16                                                   |
| Earnings Before Interest and Tax           | 19,75,000                                    | 21,50,000                                            |
| Interest Cost                              | 2,15,000                                     | 2,15,000                                             |
| Depreciation of Property & Software        | 1,75,000                                     | 1,50,000                                             |
| Depreciation of Plant & Equipment          | 2,00,000                                     | 1,80,000                                             |
| Amortisation of Patent Rights<br>Purchased | 20,00,000                                    | 20,00,000                                            |

On 1<sup>st</sup> January 2015, Alpha paid Rs 20,000,000 to acquire the right to manufacture patented components. The patent rights are purchased for 20 years but the amortization

charge for 2015 and 2016 was based on the assumption that Alpha will manufacture this component for 10 years. The management now think that there may not be a major change in technology and therefore there may be a strong case for revising the useful life of patents from 10 years to 15 years with effect from 1<sup>st</sup> January 2015.

The depreciation for Property and Software, and Plant and Equipment is based on the expected useful lives of individual assets and is in line with competitors in the industry.

The directors were concerned that the share price would decline if they published the financial statements for the year ended 31<sup>st</sup> March 2016 as they are presently drafted. However, revising the estimated useful life of the patent rights would enhance reported earnings and so the share price should increase.

- i) Calculate the EBITDA (Earnings Before Interest, Tax, Depreciation and Amortisation) for the year ended 31<sup>st</sup> December 2016 and explaining why the Investment Analysts often use this figure instead of Profit After Tax as a basis of measuring financial performance. (4)
  - ii) Estimate Alpha's share price under the assumption that draft financial statements are published as is and are adjusted to reflect the life of patents as 15 years. State any additional assumptions made. [Two figures are required for the share price – using both the new and the old amortisation calculations] (8)
- [12]

- Q. 12)** A mid sized Indian IT company has won a major overseas project in the USA and expects the project to be completed over 2 years. The revenue under the project is based on the percentage completion method (i.e. payment for work shall be made on attaining milestones during the period). Historically the company is aware of time over-runs on such similar projects which can be up to 6-9 months beyond the original estimated time.

The company is looking to hedge its foreign currency risks from the project by use of futures.

You are required to clearly highlight the risks involved in the strategy to hedge risks using futures and also the advantages and disadvantages of the same. [5]

- Q. 13)** i) Beta and Gamma are 2 general insurance companies listed on the stock exchange. The shares of both companies have a face value of Rs 100 per share with following details–

|       | Listed Price (Rs) | Price/Earnings Ratio | Book Value of Net Assets (NAV in Rs) |
|-------|-------------------|----------------------|--------------------------------------|
| Beta  | 1000              | 12                   | 600                                  |
| Gamma | 800               | 20                   | 300                                  |

- a) Explain possible reasons why the Price Earnings Ratio would differ between the two companies significantly? (3)
- b) Outline reasons why the ratio of Market Price: NAV would vary significantly between the companies? (2)
- ii) Mega Limited is an Indian company listed on the stock exchange which made a pre-tax profit of Rs 25 crores in the current year and Rs 20 crores in the previous year. The

company expects to grow both revenues and profits consistently at 25% over the next 3 years.

The company's paid up equity capital is Rs 10 crores with 1 crore shares of face value of Rs 10 each. The P/E ratio of the company is currently 18 compared to the previous year's figure of 16. The Board of the company has announced a dividend of 80% for the current year which is the same as the previous year.

Assuming an effective tax rate of 20%, calculate the dividend yield ratio and the dividend cover ratio for Mega Limited for the current year.

(5)  
[10]

- Q. 14)** i) A company's accounts should comply with various accounting concepts. Outline the cost concept and the going concern concept. (2)
- ii) A company has entered into an unusual transaction that is not directly covered by any accounting standard. Explain how the Board of Directors can go about selecting the appropriate accounting policy for this situation. (3)
- iii) The Directors of a listed company believe that if optimistic accounting policies are applied consistently, the stock market will reward the company by increasing its share price. Explain why this may not be true. (3)
- [8]

**Q. 15)** ABC Ltd.'s equity as at 31<sup>st</sup> March 2015 was as follows –

|                                  | Amount (Rs in Millions) |
|----------------------------------|-------------------------|
| Share Capital (Rs 100 per share) | 300                     |
| Securities Premium Account       | 150                     |
| Retained Earnings Reserves       | 100                     |
| Revaluation Reserves             | <u>50</u>               |
| Total Equity                     | <u>600</u>              |

During the financial year 2015- 2016, the following occurred, all of which is reflected in the statement of financial position as at 31<sup>st</sup> March 2016 –

- I.** Payment of a dividend for 2014-15 of Rs 50 Million in May 2015 (unapproved at 31<sup>st</sup> March 2015 and approved in May 2015). This dividend was subject to a dividend distribution tax of 12% payable by the company.
- II.** A 1-for-4 rights issue at Rs 175 per share in December 2015
- III.** The company made a profit before tax of Rs 75 million for the year 2015-16 including exceptional gains of Rs 10 million.
- IV.** The machinery (which had a cost price of Rs 5 million and had accumulated depreciation to date of Rs 2.4 million) was revalued at Rs 3.75 million.

Prepare the company's statement of the changes in equity for the year ended 31<sup>st</sup> March 2016 assuming an effective corporate tax rate of 25%.

[5]

- Q. 16)** Define each of the following terms used in finance: (2)
- i)** Demonetization (2)
  - ii)** Beta (2)
  - iii)** Recourse and Non recourse factoring. (2)
- [6]**
- Q. 17)** **i)** In theory, companies exist to maximize the wealth of their owners. (4)
- Explain the problems associated with demonstrating that companies actually do operate in such a way as to maximise their shareholders wealth. (4)
- ii)** The stock of XYZ Ltd is presently trading at Rs 70. The current dividend per share is Rs 10. (4)
- The company is assumed to grow sustainably at the rate of 8%. Currently the risk free rate is 6% and market risk premium is 5%. (4)
- If systematic risk is increased by 80% calculate the effect on the share price. (4)
- [8]**
- Q. 18)** A company wants to produce any one of two products, A or B. Using a probability tree, suggest which product has the higher expected value. (4)
- Product A: It will cost Rs 100,000 to develop, and if it succeeds the Business Analysis department says it will generate revenue of Rs 1,000,000. The chances of success is 50%
- Product B: It will only cost Rs 10,000 to develop. The analysts expect the product to generate Rs 300,000 in revenue. The Product has an 80% chance of success. (4)
- [4]**
- Q. 19)** **i)** Briefly list the main roles of regulation in the financial system. (2)
- ii)** Identify the regulatory bodies which regulate the Indian Financial system. (2)
- iii)** Identify the regulatory authorities currently responsible for supervising each of the following: (4)
- a)** Banks
  - b)** Superannuation Products
  - c)** General Insurance companies
  - d)** Stock brokers
  - e)** Mutual Funds
  - f)** Market for listed securities
  - g)** Foreign exchange dealers
  - h)** Money changers (4)
- [8]**
- Q. 20)** **i)** Describe the role of Investment banks. (2)
- ii)** Describe the financial services offered by an Investment Bank. (8)
- iii)** Explain the sources and application of funds for an Investment Bank (4)
- [14]**

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